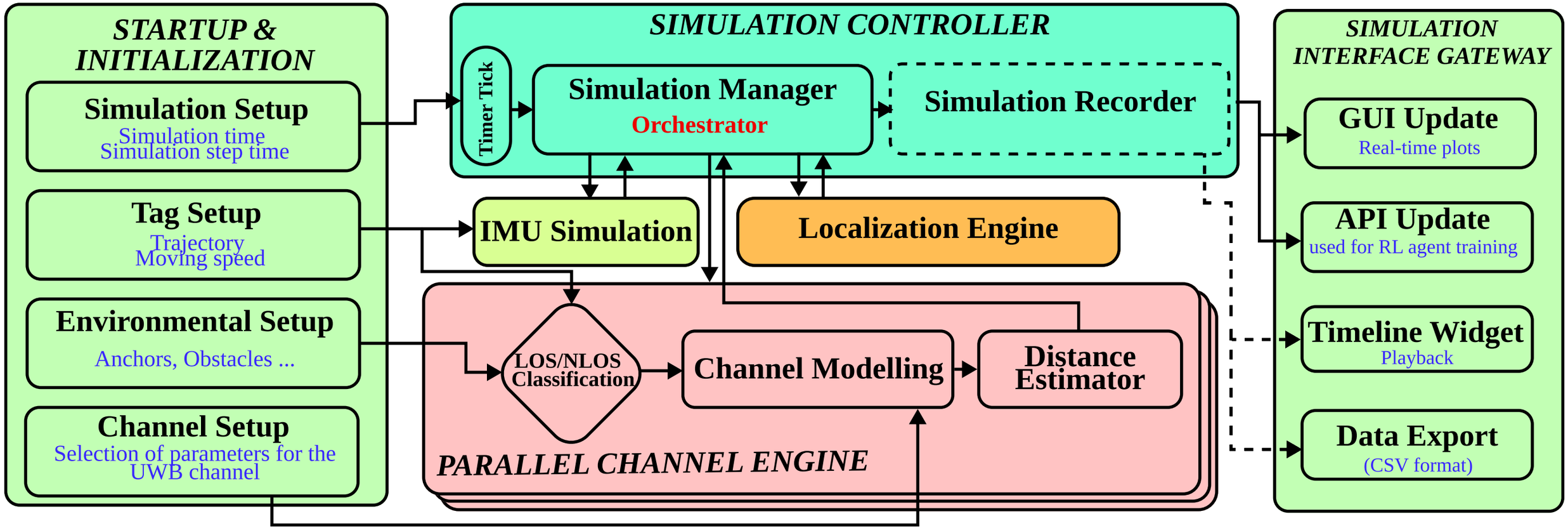
STARTUP & INITIALIZATION
Simulation Setup
Simulation time
Simulation step time
Tag Setup
Trajectory
Moving speed
Environmental Setup
Anchors, Obstacles ...
Channel Setup
Selection of parameters for the UWB channel
SIMULATION CONTROLLER
Timer Tick
Simulation Manager
Orchestrator
Simulation Recorder
IMU Simulation
Localization Engine
LOS/NLOS Classification
Channel Modelling
Distance Estimator
PARALLEL CHANNEL ENGINE
SIMULATION INTERFACE GATEWAY
GUI Update
Real-time plots
API Update
used for RL agent training
Timeline Widget
Playback
Data Export
(CSV format)
: Real-Time Data Flow
: End-of-Simulation Reporting / Export